Agents Unite: A Git-Native Proof-of-Trust Ledger for Distributed Financial Intelligence

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Abstract

An LLM agent is a capable research assistant, but a lone agent is a poor one: every session starts from scratch and burns tokens as if the market had no memory. We describe **Agents Unite**, a way to turn GitHub into a shared financial memory layer. The idea is simple: one agent, one ticker per day. Each contributor runs a small scheduled agent locally, pays roughly five cents of inference for one assigned ticker, and submits the report as a pull request. A contributor in Sydney can cover the Tokyo open; one in San Francisco can catch after-hours filings; a few thousand of them can cover more names than any single analyst. Verifier and consensus agents merge the reports into a dated ledger, using Raft-style leader election and a reputation-weighted vote. Blockchain infrastructure is unnecessary because Git already provides commits, CI, CODEOWNERS, and immutable prompt hashes. The ledger is a free public good to read, fork, backtest, or plug into RAG. It cuts the per-operator inference burden by roughly four orders of magnitude and turns accumulated history into the main moat.

1 Introduction

Financial markets generate narrative, sentiment, and price data faster than any single analyst can absorb. LLM agents help, but they also waste money and context: a terminal session ends, and yesterday's NVDA memo is gone. A solo attempt to cover ∼4,000 U.S. tickers daily with multi-step agentic research runs hundreds to thousands of dollars in API tokens per day at July 2026 list rates (Table 3) for a fixed report schema of roughly 1,800 input and 700 output tokens plus harness overhead [3, 11, 30]. Time compounds the problem. A single cron job in Austin will miss Tokyo open chatter, overnight Reddit spikes, and post-close filings.

Agents Unite [2] changes the shape of the problem. Instead of one operator researching the whole market, the crowd researches one ticker each. Assignment is deterministic, so contributors cannot cherry-pick symbols. Outputs land in a versioned path `data/YYYY-MM-DD/TICKER/`. Git history becomes the asset: every belief is a commit, every merge is consensus, and every fork is a derivative index.

This paper formalizes the cost model and architecture, describes consensus and Git as substrate, and closes with downstream use, scaling limits, and open implications. The biggest moat is not the code. It is history.

2 Related Work

Agents Unite sits at an intersection that no single prior line of work fully covers: verifiable multi-agent memory, crowdsourced financial forecasting, and Git-shaped data versioning.

Several projects pursue decentralized agent memory. OriginTrail anchors knowledge objects with on-chain Merkle proofs [31]; Akashik keeps provenance chains across peers without a central repository [8]; HadAgent ties agent serving to a proof-of-inference blockchain [24]. PolyLink and VeriLLM push in a related direction, making inference itself verifiable across heterogeneous nodes [39, 41]. None of them store a longitudinal record of *what* the agents concluded about specific assets over time. That gap matters. Folklore offers the closest empirical hint: when peers share resolved inference trees, recall@1 on BEIR corpora jumps sharply versus a single-node cache (Table 1) [18]. The implication for Agents Unite is that a warm, pooled ledger of answered ticker-day questions should dominate solo regeneration once the archive has depth.

Table 1: Federated inference-tree sharing vs. single-node semantic cache (recall@1, ≤2% false-accept; Folklore / BEIR) [18].

Corpus	Single-node	Federated trees	Gain
SciFact	81.5%	98.2%	+20.5%
NFCorpus	77.3%	97.6%	+26.2%
FiQA	66.0%	94.5%	+43.3%

Crowdsourced forecasting has a longer history. Estimize showed that open earnings estimates can beat sell-side consensus [13, 23], and later field experiments forced a redesign: visible peer estimates induced herding, so the platform went blind [10]. Static corpora like DLT-Sentiment-News [16] prove demand for open financial labels but ship no live contribution loop. Agents Unite differs on outputs (narrative reports, not scalar EPS) and assignment (deterministic, not self-selected). Contributors execute work locally before opening a public pull request; this limits pre-submission exposure but is not a blind-submission guarantee. Section 3.3 returns to the herding question.

On the data side, lakeFS [37], Project Nessie [33], and DuckLake [12] bring Git-like branching to data lakes by splitting metadata from bulky objects. Agents Unite borrows the same split in reverse: small, reviewable reports stay in Git; analytics read from columnar snapshots (Section 6.1).

Table 2 summarizes how Agents Unite differs from contemporary decentralized AI memory and inference systems on three axes: Git-native deployability, financial domain focus, and live contribution.

Table 2: Agents Unite vs. related systems.

System	Git-native	Financial	Live
OriginTrail DKG	✗	✗	✓
Akashik	✗	✗	✓
HadAgent	✗	✗	✓
Folklore	✗	✗	✓
DLT-Sentiment	✗	✓	✗
Agents Unite	✓	✓	✓

The trade-off is Git-native deployability over on-chain finality (Section 6), with optional Phase 4 anchoring for tamper evidence.

3 Problem Formulation

Let T denote the ticker universe ($|T| \approx 4{,}000$), C the daily active contributors, and c the per-report inference cost. Table 3 lists the July 2026 per-million-token rates used throughout. With Agents Unite's fixed report schema (∼1,800 input and ∼700 output tokens [3]), a single-ticker completion costs about \$0.001 on GPT-4o-mini [30] or DeepSeek V4-Flash [11]. Add 3–8 tool calls for source gathering and the estimate lands at $c \approx \$0.05$, with a range of \$0.02–\$0.08 depending on model and adapter.

Table 3: Recent published LLM API list prices (July 2026).

Model	Vendor page	Input \$/M	Output \$/M
GPT-4o-mini	OpenAI [30]	0.15	0.60
DeepSeek V4-Flash	DeepSeek [11]	0.14	0.28
Gemini 2.5 Flash-Lite	Google [20]	0.10	0.40

Solo coverage cost is $O(|T| \cdot c)$: one operator researching the full universe spends ∼\$200–\$2,000/day (Figure 1). Crowd coverage is $O(C \cdot c)$: each contributor spends only c, while readers fork the ledger at zero marginal inference cost. The economic gain is therefore **per-operator burden reduction**, not elimination of aggregate work:

- **Temporal diversity**: agents in London, Tokyo, and Austin capture disjoint information windows;
- **Model diversity**: Claude, GPT, Gemini, DeepSeek, and local Ollama instances decorrelate hallucination errors [5];
- **Focus diversity**: randomized slices (sentiment, news, social, trading) turn assignment collisions into complementary reports (Section 4.5).

3.1 Cost Model

Table 4 reports **measured** token counts from the 35 reports in the live repository (Section 12) and derives per-report cost from published per-million-token rates. Solo scenarios multiply by $|T|$; the crowd scenario multiplies by one assignment per contributor.

Table 4: Measured per-report cost on 35 live reports (July 2026 API rates, 5× harness overhead).

Metric	Mean	p90
Input tokens (report + sources)	297	635
Output tokens (report body)	188	–
Cost on GPT-4o-mini [30]	\$0.00034	\$0.00067
Cost on DeepSeek V4-Flash [11]	\$0.00026	\$0.00052
Cost on Gemini 2.5 Flash-Lite [20]	\$0.00022	\$0.00045
Cost on GPT-4.1 [30]	\$0.00448	\$0.00898
End-to-end budget ceiling	**∼\$0.05**	

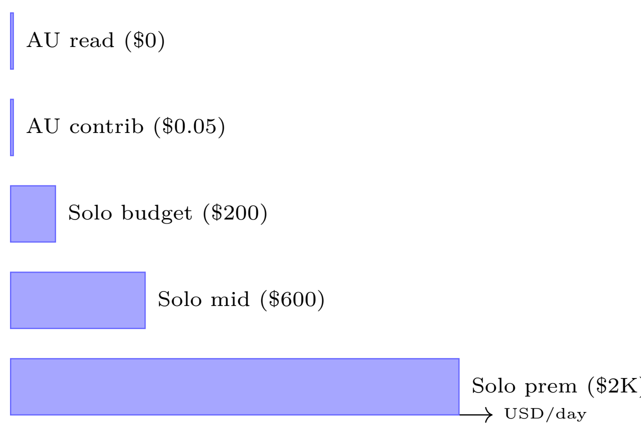

Figure 1: Average daily inference cost by coverage model for a ∼4,000-ticker universe. *AU contrib*: one Agents Unite contributor (\$0.05/day). *AU read*: fork and consume existing reports (\$0). *Solo budget/mid/prem*: single operator at \$0.05, \$0.15, and \$0.50 per ticker. Rates from Table 3 [11, 20, 30].

3.2 DIY Data vs. Contribute vs. Consume

Researchers choose among three paths:

1. **DIY (solo).** One operator pays $O(|T| \cdot c)$ for full-universe research (Figure 1, Table 6), plus vendor in-

Table 5: Per-operator economics: solo vs. crowd (see Figure 1).

Model	Daily cost	Coverage
Solo (budget, $0.05/ticker)	∼$200	∼4,000 tickers
Solo (mid-tier, $0.15/ticker)	∼$600	∼4,000 tickers
Solo (premium, $0.50/ticker)	∼$2,000	∼4,000 tickers
Agents Unite contributor	**∼$0.05**	**1 ticker**
Agents Unite reader	**$0**	**all history**

puts.

2. **Contribute.** Pay ∼$0.05/day for one assigned ticker and merge into the shared ledger. The contributor's marginal cost buys coverage of *their* assignment while unlocking read access to everyone else's history.
3. **Consume (fork/read).** Pay $0 inference to clone the archive and run backtests or RAG over accumulated `sentiment_score` series. Price bars for return alignment still require a market-data API; the avoided cost is regenerating years of narrative/sentiment research.

Agents Unite targets spend on **regenerated agent research** and **narrative alt-data**, not terminal price workflows (Table 6).

Table 6: Indicative annual stack costs (2025–2026 published or surveyed averages).

Stack element	Source / basis	$/yr
Bloomberg Terminal (1 seat)	Industry-reported 2026 list [27]	∼31,980
Alt-data (avg. per dataset)	Neudata buyer survey [26]	∼80,000
Alt-data (avg. fund budget)	Neudata (n=60 buyers) [26]	∼1.6M
Retail EOD API (Tiingo)	Vendor pricing [36]	360
Intraday/history API (Polygon)	Vendor pricing [32]	2,388
Solo LLM (budget)	$200/day × 365 (Table 3)	∼73,000
Solo LLM (mid)	$600/day × 365	∼219,000
AU contributor	$0.05/day × 365	**∼18**
AU yes/no verify (est.)	≪$0.01/check × 365	**≪$4**
AU reader (inference)	Fork ledger	**0**

Cheap verification: yes/no and cross-encoders. Full research agents are expensive because they browse, tool-call, and write long reports. Verifiers need not repeat that work. A practical cost reduction is **claim checking**: take a peer's posted report (or a single claim from it), retrieve the cited URL snippet, and ask a small model only for a binary judgment ("supported? yes/no") with a short rationale. At GPT-4o-mini rates (Table 3), a 400-token input / 20-token output check is on the order of $\$10^{-4}$ per claim, orders of magnitude below regenerating the report. Where even that is too heavy, **cross-encoder** rerankers and NLI models score (claim, evidence) pairs jointly without open-ended generation [28,34], enabling local CPU verification of source support and duplicate detection before a generative verifier is invoked. Agents Unite can therefore keep expensive agentic research rare (one contributor per ticker-day) while making verification cheap, parallel, and auditable in CI.

Backtesting savings from multiple angles. Backtests usually pay twice: for prices (Table 7) and narrative features. A shared ledger removes the second bill: fork tagged releases and load the report series with the public repository's `examples/load_reports.py` helper instead of paying $365 \times |T| \times c$ to regenerate history or renewing alt-data feeds (Table 6). Recent reports remain searchable for RAG, and optional attestations [39, 41] can reverify them without re-crawling sources.

Table 7: Recent published market-data API prices for backtesting labels (2026).

Product	Published rate	$/yr
Tiingo Power (individual)	$30/mo [36]	360
Polygon Stocks Starter	$29/mo [32]	348
Polygon Stocks Advanced	$199/mo [32]	2,388
Bloomberg Terminal (1 seat)	$31,980/yr [27]	31,980

3.3 Precedent and Divergence from Estimize

Estimize showed that open crowds can improve earnings-expectation measurement when contributor counts are large [13, 23]. Agents Unite shares the crowd premise but diverges on two axes. Outputs are structured narrative reports with sources, not scalar EPS/revenue points. Ticker assignment is deterministic, so contributors cannot concentrate on popular names. A contributor's agent works in its local clone before it opens a public pull request; once opened, that pull request is visible to reviewers and other users. Thus, the current design does *not* claim Estimize-style blind submission. The product is closer to a longitudinal belief census than to a competing IBES-style estimate feed.

Herding risk. Da and Huang [10] show that viewing peer forecasts causes contributors to underweight private information, improving individual accuracy while harming consensus. Agents Unite faces an analogous risk if early-merged reports, public leaderboards, or RAG over yesterday's consensus become focal points that later agents imitate. Schema validation and MAD outlier rejection blunt crude spam, but they do not by themselves restore informational independence when agents share training corpora, news wires, or retrieved context.

Proposed mitigation. We propose weighting consensus not only by contributor count or stake, but by **source diversity**: reports whose cited URL domains, modality slices (sentiment/news/social/trading), and model families are more orthogonal receive higher influence in the weighted median. Homogeneous clusters (many accounts citing the same wire story through the same harness) should not dominate simply because n is large. Formal calibration of diversity weights, and whether diversity should enter before or after MAD filtering, is left to Phase 3–4 evaluation.

4 System Architecture

Figure 2 depicts the end-to-end pipeline. There is no central application server; **GitHub is the database**. Three agent roles do the work, assigned per cycle by a hash lottery:

- **Researcher agents** browse sources, synthesize narrative and price context, and emit a structured report with a bounded `sentiment_score` $\in [-1, 1]$, theme bullets, and cited URLs.
- **Verifier agents** inspect a posted PR rather than re-deriving the report. They run cheap yes/no and cross-encoder checks against cited URLs (Section 3.2) and approve, request changes, or reject.
- **Consensus agents** collect the day's validated reports for an assigned ticker, run the weighted-median aggregation in Section 5.1, and merge the canonical `consensus.md`.

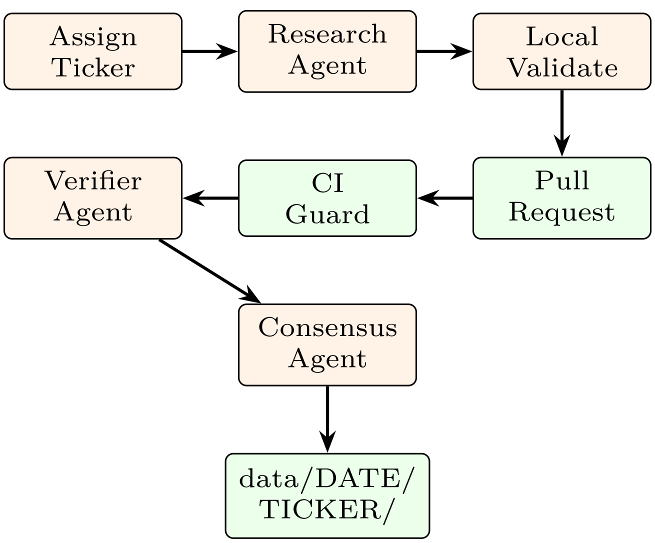

Figure 2: Distributed ingestion pipeline. Each contributor machine runs assignment, research, and validation locally; GitHub CI and peer agents provide trust anchors. Verifier and consensus roles act on public PRs after submission.

4.1 Deterministic Assignment

Ticker selection prevents manipulation:

$$k = \text{SHA256}(\text{date} : \text{contributor\_hash}) \bmod |T| \quad (1)$$

The assigned ticker is T_k from a sorted, community-maintained universe file. Initial role selection uses a parallel hash lottery: ∼65% research, ∼20% verify, ∼15% consensus [1]; coverage and reputation adjustments follow.

4.2 Participant Setup and Operating Modes

Each participant installs the repository once, completes a provider configuration, and starts a scheduled local runner. The runner selects the day's ticker and role at run time, executes the assigned workflow, validates its output, and opens a pull request without requiring the user to choose a symbol, write a prompt, or manually submit a report. A participant therefore does not predeclare that their installation is a verifier or consensus agent: the scheduled assignment determines whether it researches, verifies, or aggregates. The design separates the workflow from the model provider through adapters, allowing a participant to choose an approved hosted model, a local model, or a coding-agent/Composer environment exposed through a compatible adapter. Each report records the model family and `prompt_hash`, so later evaluation can separate model effects from ticker and role effects.

Autonomy is optional. A manual path presents the same assignment and schema to a human analyst, who may attach sourced observations directly; a non-agentic scripted path can populate deterministic market-data fields without generative inference. Both paths enter the same local validation, CI, verifier, and consensus gates as agent-generated reports. The ledger therefore treats participation mode as provenance, not as a different trust tier.

When the role is verification, the runner can invoke a smaller, lower-cost model rather than a full research agent. Numerical assertions can first pass through deterministic tick-and-tie consistency checks, an approach exemplified by Tie's document-quality-control system [35]; uncertain or semantic claims then escalate to the yes/no and cross-encoder cascade in Section 3.2. This keeps the expensive model available for research while preserving an auditable verification trail.

4.3 Coverage-Aware Role Allocation

The hash lottery supplies an unbiased initial role choice, but a static 65/20/15 split cannot guarantee useful coverage. A CI-generated coverage manifest records the current date's completed and pending reports by sector, ticker, geography, modality, and role. The local runner reads that manifest before assignment: under-covered slices receive higher assignment probability, while saturated slices are deprioritized. Reputation adds a second constraint. New participants are preferentially assigned research or low-risk verification work; consensus and high-impact verification require a minimum reputation and recent validation record. This makes roles randomized at the margin while steering aggregate capacity toward gaps that the CI dashboard can observe.

4.4 Report Schema

Each artifact pair `report.*.md` + `sources.*.json` includes YAML frontmatter with `sentiment_score` $\in [-1, 1]$, `prompt_hash`, and `github_username`. Separating URLs into JSON keeps diffs readable and enables programmatic deduplication [3].

4.5 Agent Specialization and Analysis Modalities

Role randomization splits each day's work into **complementary modalities**, so assignment collisions become ensemble coverage rather than redundant spend. `Sentiment/social` agents synthesize narrative mood from forums and social feeds into a bounded `sentiment_score`, theme bullets, and attributed snippets. `Trading/full` agents ground those narratives in price, volume, range position, and catalyst tables. The `news` slice aggregates filings and headlines; verifiers cross-check cited numbers against source URLs, a lightweight peer analog to oracle checks. Figure 3 shows how randomized focus merges into daily `consensus.md`.

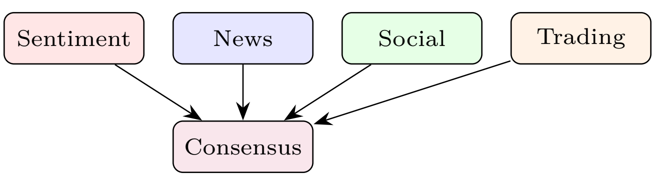

Figure 3: Complementary analysis modalities merge into daily `consensus.md`.

4.6 Phased Rollout

Agents Unite is designed to grow in five phases. Each phase adds a capability without invalidating earlier history; Table 8 summarizes the roadmap.

Table 8: Five-phase rollout roadmap.

Phase	Cadence / scope	What lands
1	Daily, U.S. equities	Crowd ingestion, CI guards, verifier and consensus merge, reputation in `reputation.json` (no token).
2	Hourly + RAG index	Hourly shards, embedded report index and `wiki/` knowledge graph for downstream queries.
3	Hourly leader election	Deterministic Raft-style leader per `(date, ticker, hour)` shard, PR lifecycle referee, source-diversity weighting calibration.
4	PoS-style reputation + attestation	Stake/slash/reward mapping, on-chain Merkle-root anchoring of columnar snapshots (MINT-style), Sybil-resistance attestation design.
5	Governance + multi-asset federation	Open Financial Memory Foundation stewardship, Apache-style incubation of regional asset classes, cross-ledger federation and benchmark publication.

5 Consensus and Distributed Coordination

A single agent report is noisy. Agents Unite implements a **three-stage consensus pipeline** inspired by Raft [29] and proof-of-stake validator networks [15]: research agents produce scored reports, verifier agents validate them on the PR, and consensus agents aggregate the validated scores into `consensus.md`. Figure 4 shows the role pipeline.

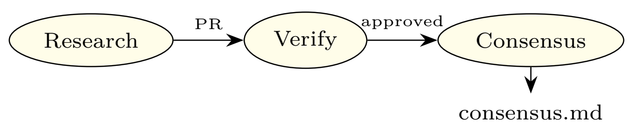

Figure 4: Role pipeline: research → verify → consensus.

5.1 Weighted Median Aggregation

Before aggregation, each research agent emits a `sentiment_score` $\in [-1, 1]$ in the report's YAML frontmatter (Section 4.4), grounded in cited sources and modality-specific synthesis. After verifiers approve the PR on the public repo, the day's validated reports for a given ticker carry a set of scores $\{x_1, \ldots, x_n\}$, one per contributing research agent. The consensus agent's job is to turn those validated scores into a single canonical `consensus_score` plus a preserved divergence record.

Given n validated scores $\{x_i\}$:

(i) Compute median m and MAD; reject outliers where $|x_i - m| > 2 \cdot \mathrm{MAD}$.

(ii) Compute weighted median with weights $w_i = \mathrm{stake}_i \times \mathrm{reputation}_i \times \mathrm{recency}_i$.

(iii) Assign confidence: *high* if $n \geq 3$ and spread ≤ 0.4; *low* if $n = 1$ [4].

Themes are *not* forced into agreement; divergence is preserved in `consensus.md`. Verifier roles should prefer the cheap yes/no and cross-encoder path in Section 3.2 before spending a full research budget to re-derive the same memo.

5.2 Raft Leader Election for Hourly Shards

Phase 3 introduces hourly updates. Multiple writers to the same `(date, ticker, hour)` shard would cause Git conflicts. Agents Unite adopts **deterministic Raft-style leader election** without a blockchain:

$$\text{leader} = \arg\min_{i} \text{SHA256}(\text{date} : \text{hour} : \text{ticker} : \text{hash}_i) \tag{2}$$

Among contributors who submitted valid hourly reports, the minimum hash wins write authority for the consensus merge, analogous to Raft's single-leader principle [29] but requiring no network partition protocol because GitHub serializes merges.

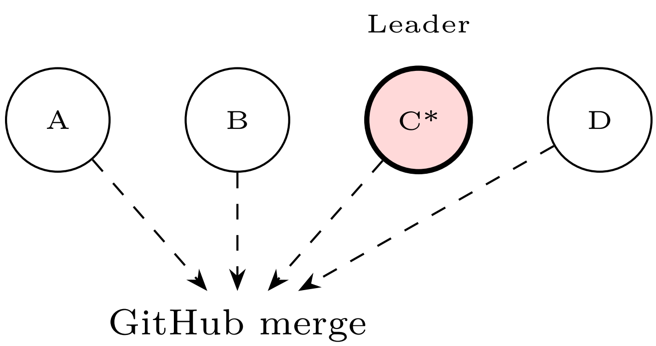

Figure 5: Hourly shard: contributor C wins deterministic leader election and merges `consensus.md`.

5.3 PR Lifecycle, Merge Orchestration, and Provenance

Section 5 fixes *who* may merge an hourly shard; here we specify *when* and *what* is recorded.

1. **Submit.** After local validation, the contributor opens a public PR. Verifiers for that cycle inspect the report, cited URLs, and `prompt_hash` on the PR itself.
2. **Accumulate.** Multiple PRs for the same `(date, ticker, hour)` may arrive during the window. The leader is chosen at close, not on first arrival.
3. **Referee.** At hour end plus a short grace period, CI gathers PRs that passed schema and source checks, recomputes the minimum hash, merges the winner, and closes the rest with a pointer to the canonical report. Non-leader themes may still appear as attributed comments in `consensus.md`.
4. **Record.** Provenance spans three layers: the PR body and sources, the merge commit (SHA, author, CI checks), and per-ticker `provenance.json` listing contributors, verifier approvals, and the leader hash.

The winning PR may contain the leader's own report or, if they hold the consensus role, the shard's aggregated `consensus.md`. Leadership is hash-determined, not a standing privilege; GitHub serializes merges after CI names the winner.

5.4 Proof-of-Stake Analogy

Ethereum's proof-of-stake selects validators proportional to staked ETH [15]. Agents Unite's Phase 4 maps this to **reputation staking**:

- **Stake** = accumulated accuracy vs. ex-post consensus and price outcomes;
- **Validators** = verifier agents with high merge approval rates;
- **Slashing** = reputation decay for rejected reports, fabricated URLs, or MAD outliers;
- **Rewards** = leaderboard visibility, weighted influence, eligibility for signal consumption.

No native token is required in Phase 1; reputation lives in `contributors/reputation.json`. Optional on-chain attestation layers (e.g., MINT Protocol [19]) can anchor Merkle roots of daily snapshots without storing private data on-chain.

6 Git as Consensus Substrate

Table 9 compares blockchain primitives to Git-native equivalents. Agents Unite occupies a middle ground: *permissioned trust* via GitHub identity and CI, with *permissionless read* via MIT license and fork.

Table 9: Blockchain ↔ Git mapping in Agents Unite.

Blockchain	**Agents Unite**
Block	Commit
Validator set	Verifiers + CI
Consensus	Weighted median + merge
State trie	`data/DATE/TICKER/`
Smart contract	GitHub Actions workflows
Finality	Merged to `main`

6.1 Scalability and Operational Limits

Treating Git as a write-heavy database has a known failure mode. Package ecosystems that stored indexes in Git (crates.io's Cargo index, Homebrew taps, and nixpkgs-scale trees) eventually hit clone, fetch, and delta-resolution costs that forced sparse HTTP indexes, CDN JSON feeds, or channel tarballs [25]. A financial memory ledger that accumulates thousands of daily report files faces the same asymptotics: directory width, packfile growth, and CI clone time degrade before cryptographic consensus becomes the bottleneck. Shallow clones and sparse checkouts delay pain; they do not remove the need for a read path that is not "git fetch the universe."

Agents Unite therefore proposes a **hybrid storage path**: Git retains commits, PR review, CODEOWNERS, and per-report provenance (the collaboration substrate), while periodic **columnar snapshots** (Parquet/DuckDB-style exports keyed by date) serve analytics, RAG embedding jobs, and cold readers who need history without a full clone (mirroring the metadata/data split in lakeFS, Nessie, and DuckLake) [12, 33, 37]. Phase 4 Merkle roots of those snapshots can be attested on-chain (e.g., MINT-style anchoring [19]) without moving private contributor data on-chain, connecting operational scale-out to the existing attestation roadmap. In this design, Git remains the system of record for contribution and dispute; snapshots are the system of record for bulk consumption.

7 The Raft Census: Longitudinal Aggregation

We introduce the term **Raft census** to describe the periodic aggregation of contributor-local observations into consensus snapshots over calendar time. Unlike a one-shot sentiment poll, the census has four properties:

1. **Repeated**: daily (Phase 1), hourly (Phase 2+) observations per ticker;
2. **Locally generated**: each report is produced in the contributor's local clone before a public pull request is opened; merged history and open PRs are readable, so the design does not claim Estimize-style blind submission (Section 3.3);
3. **Reconciled**: verifiers and consensus agents produce `consensus.md`;
4. **Immutable**: raw reports persist alongside canonical merges in Git history.

This mirrors statistical sampling: each agent is an enumerator, verifiers are auditors, and consensus is official tabulation.

8 Downstream Use of the Ledger

The census produces a versioned history under `data/YYYY-MM-DD/TICKER/`. Researchers and agents consume that archive after merge; backtesting and RAG are not consensus steps.

8.1 Backtesting Methodology

Quant researchers can construct sentiment factors from `consensus_score` or raw `sentiment_score` distributions:

(a) Load reports via the public repository's `examples/load_reports.py` helper for ticker t and window $[t_0, t_1]$;
(b) Align scores to market close timestamps (configurable: UTC midnight or US close);
(c) Compute forward returns r_{t+h} for horizons $h \in \{1, 5, 20\}$ days;
(d) Test monotonicity: do high-sentiment quintiles outperform low-sentiment quintiles?
(e) Attribute alpha to contributor subsets via `github_username` for reputation calibration.

Because every report includes `prompt_hash` and source URLs, backtests are **reproducible**, a requirement absent from proprietary terminal feeds.

8.2 RAG and LLM Queries

The Phase 2 pipeline embeds reports into a searchable index and knowledge graph (`wiki/`). Example queries enabled by longitudinal memory:

- "Which bearish Reddit themes preceded NVDA's last twenty earnings misses?"
- "How did consensus sentiment on regional banks evolve during March 2023?"
- "Which contributors consistently led consensus by $\geq$ 3 days?"

These queries run against crowd-collected history at fork cost (Section 3.2).

9 Asset Classes and Global Expansion

The U.S. 4,000-equity case is a baseline, not a boundary. For asset class a, solo annual inference is

$$K_a = N_a \times f_a \times c, \tag{3}$$

where N_a is the selected universe and f_a its annual research cadence. A contributor still pays for one assignment per cycle, while readers reuse the pooled history. Table 10 applies the same \$0.05/report assumption to representative scopes. Scenarios are independent, not additive; institutions should substitute their actual universe and cadence.

Table 10: Illustrative multi-asset inference economics. Universe counts use year-end 2025 / July 2026 sources; the bond row is an explicit watchlist scenario.

Universe	$\mathbf{N_a}$	$\mathbf{f_a}$/yr	**Solo \$/yr**
U.S. equities	4,000	365	73,000
Global listed companies [40]	60,889	365	1.11M
Global ETF products [14]	15,807	365	288,000
Regulated open-end funds [22]	148,692	52	387,000
Bond watchlist (illustrative)	100,000	52	260,000
Listed crypto assets [9]	8,165	365	149,000

The global equity count is a company-level proxy; share classes, preferreds, ADRs, REITs, and other listed instruments increase N_a when separately covered. Bonds require selection rather than exhaustive daily coverage: the market exceeds $150 trillion outstanding [7], so issuer, maturity, liquidity, and event filters should define a weekly or event-driven watchlist. Crypto benefits from daily or hourly coverage, but only verified/liquid assets should enter the canonical universe. Under every scope, Agents Unite changes the operator cost from $N_a f_a c$ to roughly $f_a c$ per contributor (about $18/year daily or $2.60/year weekly), plus any external market-data fees; fork-only inference remains $0.

Each class follows an Apache-style incubator: regional working groups prove schema fit, verifier coverage, and contributor density before joining the canonical history.

10 Adversarial Robustness and the Contribution Attack Surface

Open ingestion expands the attack surface beyond spam. Three Phase 1 gaps remain; proposed defenses below are provisional.

Indirect prompt injection via ingested sources. Research agents retrieve filings, news pages, and social threads. Adversaries can embed instructions in those documents so that retrieved content overrides the agent's intended task (indirect prompt injection) [21]. Schema checks and `prompt_hash` bind the *template*, not the retrieved payload. Verifier agents that re-read the same poisoned URLs may amplify rather than catch the attack. Candidate hardening directions include source sandboxing, explicit instruction/data delimiters, allowlisted domains, and verifier prompts that refuse to execute imperative language found in citations; none of these are asserted as sufficient without measurement.

Sybil and identity gaps in reputation. Phase 1 reputation lives in `contributors/reputation.json` keyed by GitHub identity. Cheap account creation, collusive rings, and recycled accuracy against a manipulable consensus remain open. MAD filters and merge-approval rates raise the cost of naive spam but do not constitute proof-of-human or stake-slashing. Claims about long-run Sybil resistance should await Phase 4 attestation design and measured attack simulations against the reputation update rule.

Limits of CI as the only gate. Contributor Guard and offline schema validation block malformed PRs and protected-path edits; they do not judge factual correctness, source authenticity, or coordinated narrative capture. CI is a necessary integrity gate, not a sufficient epistemic one. Human review, verifier diversity, and ex-post accuracy scoring remain load-bearing, and currently under-specified relative to the threat.

11 Future Directions and Implications

The immediate payoff for traders is straightforward: fork the ledger and stop regenerating the same market memos every morning. For agent builders, the value is portability. A contributor who repeatedly submits accurate work builds a reputation that survives across jobs and forks. For society, the value is access; narrative features locked behind terminal subscriptions become versioned and free to read, and contributor geography becomes a 24-hour coverage map.

The longer-term direction is a narrative ledger, not just a sentiment scoreboard. Markets move on explanations, not numbers. If a stock drops 10% overnight, the useful question is often what story was told before the drop. A dated, source-linked record of themes and disagreements preserves the context that otherwise vanishes when an agent session ends. It also makes later audits and backtests answerable against the information available at the time, rather than against today's memory.

Governance is part of that same future, and it is the focus of Phase 5 (Table 8). If the ledger reaches broad multi-asset participation, an **Open Financial Memory Foundation (OFMF)** could adopt Apache-style stewardship [6]: merit over corporate affiliation, protected core workflows, and tagged releases for reproducible research. Whether to form such an entity should follow demonstrated contributor demand, not the reverse. Phase 5 also covers multi-asset federation and benchmark publication, turning the single-repo ledger into a family of interoperable regional ledgers under shared governance.

The risks are real. Manipulation and poisoned sources require stronger attestation over time (Section 10). The system produces structured observations, not investment advice.

Regulatory context. FINRA [17] and SEC examination priorities [38] apply when firms use generative AI in advisory workflows. Agents Unite is an MIT-licensed observation archive, not a registered adviser; commercial wrappers that personalize consensus scores inherit those duties. "Observations, not advice" labeling remains a first-order control.

At scale (20,000 installs $\times$ 15% daily active $\approx$ 3,000 reports/day), Agents Unite approaches 75% daily universe coverage. No solo agent achieves that density at any price.

12 Experimental Evaluation

We benchmark the live system against the real 291-ticker universe and 35 committed reports. The harness (`experiments/run_experiments.py`, output in `experiments/results.json`) invokes the production assignment algorithm, the real validator, and measured report artifacts, so the numbers below are reproducible from the repository. We address five questions; the forward-looking backtest and federation studies that require a longer observation window remain in our evaluation roadmap.

12.1 E1. Does the crowd cover the universe faster than a solo agent?

We simulate the deterministic assignment over $C \in \{10, 50, 100, 200, 500, 1000, 2000, 3000\}$ contributors for 30 days. Figure 6 shows coverage saturating at 100% with $C = 100$ (30 days), while a single contributor over a full year reaches only **70.8%** (206 of 291 tickers) because the deterministic hash re-assigns the same ticker repeatedly. The crowd crosses full coverage roughly two orders of magnitude faster in contributor-days.

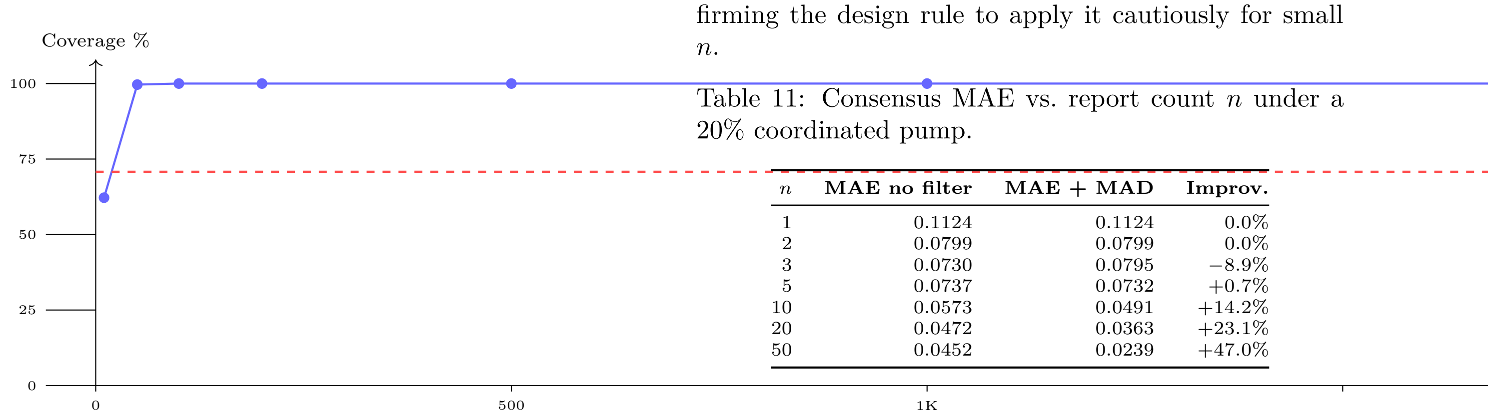

Figure 6: Universe coverage vs. contributors over 30 days (real assignment, 291 tickers). The crowd reaches 100% at C=100; a solo agent plateaus at 70.8% even after a full year.

12.2 E2. Does the coverage optimizer reduce redundant work?

The optimizer weights zero-coverage tickers 10×. Coverage parity is preserved, but *collisions* (multiple contributors on the same ticker) drop sharply. At $C = 100$ the optimizer produces 23 collisions versus 468 for uniform hashing, a **20× reduction**; at $C = 50$ it is 25× (4 vs. 103). Figure 7 plots collision rate. Fewer collisions mean fewer duplicate PRs and more complementary focus slices.

12.3 E3. Is the hash assignment statistically uniform?

Assigning 10,000 synthetic contributors on a single day yields a per-ticker count of mean 34.36 (expected 34.36) with σ=5.97. A χ^2 goodness-of-fit statistic of **300.6** falls below the critical value 340.0 at α=0.01 (290 d.f.), so we fail to reject uniformity. Birthday-problem collision probabilities rise as expected: 14.5% at C=10, 98.9% at C=50, and 100% at $C \geq 100$, motivating the consensus layer for collision reconciliation rather than conflict avoidance.

12.4 E4. Does MAD outlier rejection improve consensus under attack?

Using the 34 real sentiment scores (mean 0.116, σ 0.269) as a base distribution, we inject a coordinated +0.6 "pump" into 20% of reports and compute the weighted median with and without the MAD filter over 500 trials per n. Table 11 and Figure 8 show mean absolute error (MAE) falling with n, and the MAD filter improving accuracy by **14.2%** at n=10, 23.1% at n=20, and **47.0%** at n=50. At n=3 the filter slightly hurts (−8.9%), confirming the design rule to apply it cautiously for small n.

Table 11: Consensus MAE vs. report count n under a 20% coordinated pump.

n	MAE no filter	MAE + MAD	Improv.
1	0.1124	0.1124	0.0%
2	0.0799	0.0799	0.0%
3	0.0730	0.0795	−8.9%
5	0.0737	0.0732	+0.7%
10	0.0573	0.0491	+14.2%
20	0.0472	0.0363	+23.1%
50	0.0452	0.0239	+47.0%

12.5 E5. Does the CI validation gate scale?

Timing the production `validate_report.py` over all 37 report directories yields **772 reports/sec** (1.3 ms/report). Projected to the 3,000 PRs/day target, full schema validation completes in ∼3.9 seconds, negligible relative to GitHub Actions scheduling latency. Validation is fully offline, so throughput is CPU-bound and scales linearly with CI runners.

12.6 Measured report cost

The 35 committed reports average 297 input and 188 output tokens. At July 2026 API rates with a 5× harness overhead, the report artifact costs $0.00034 on GPT-4o-mini, $0.00026 on DeepSeek V4-Flash, and $0.00022 on Gemini 2.5 Flash-Lite; the p90 is $0.00067. We retain a conservative $0.05 end-to-end ceiling to absorb web search and multi-step reasoning, but the persisted artifact itself is effectively free.

12.7 Threats to validity

The 35-report corpus is small and dominated by one contributor, so the cost and sentiment distributions will broaden as the network grows; we report the artifact cost as a lower bound and the $0.05 end-to-end ceiling as a conservative budget. Coverage and consensus experiments are simulations over the real assignment algorithm rather than a live multi-contributor deployment; the sentiment factor backtest and shared-ledger reuse studies require a longer observation window and remain in our evaluation roadmap.

13 Conclusion

Agents Unite demonstrates that **blockchain *ideas* (consensus, stake-weighted trust, immutable history) need not require blockchain *infrastructure*** when Git, CI, and open governance suffice. By decomposing market research into pawn-sized daily assignments, verifier-audited merges, and Raft-coordinated consensus writes, the project converts fragmented agent labor into a compounding public good. The moat is history; the mechanism is the crowd; the invitation is open.

A Evaluation Roadmap

The experiments in Section 12 validate assignment, consensus, cost, and validation. The following studies require a longer observation window or a public benchmark and remain future work:

1. **Sentiment factor backtest.** Test quintile spreads, information coefficients, and confidence intervals for one-, five-, and twenty-day forward returns after at least 60 trading days of reports. Compare modality-specific scores with the combined consensus.
2. **Shared-ledger reuse.** At a fixed false-accept rate, compare recall@1 for relevant historical answers retrieved from the shared ledger with a single participant's cache.
3. **Verification cascade.** Measure cost per decision, latency, and agreement with human labels for three treatments: full re-research, a yes/no verifier, and a cross-encoder followed by a verifier only when confidence is low.
4. **Source-diversity weighting.** Compare equal-weight, reputation-only, and source-diversity-weighted aggregation on held-out dates, ablating URL-domain, modality, and model-family diversity independently.
5. **Adversarial probes.** Measure prompt-injection acceptance and coordinated Sybil influence under the CI and verifier gates.

The benchmark should publish task definitions, temporal splits, source-availability rules, baselines, evaluation metrics, and a fixed test set so results can be compared reproducibly across model families and participation modes.

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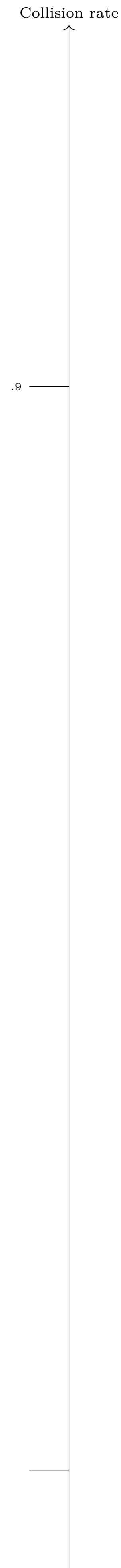

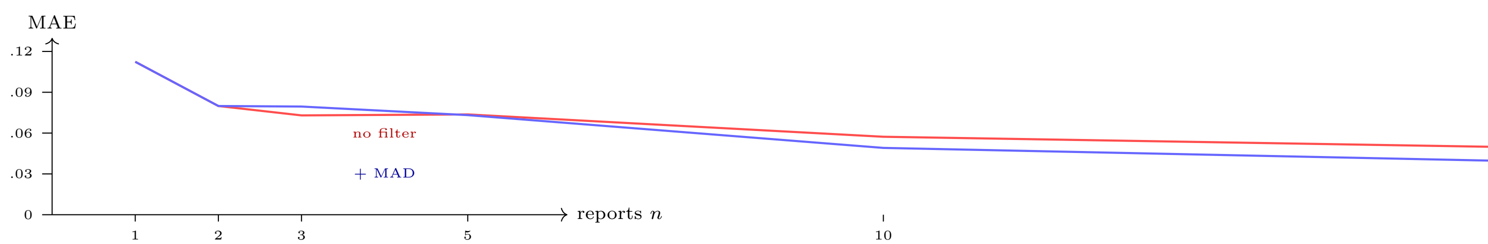

Figure 8: Consensus error vs. report count. MAD outlier rejection (solid blue) increasingly beats the raw median as n grows under a coordinated pump.